



ENGINEERING MATHEMATICS-II

Department of Science and Humanities



ENGINEERING MATHEMATICS-II

UNIT 1: INTEGRAL CALCULUS

Class 9: change of variables in
triple integral, cylindrical
coordinates

TRIPLE INTEGRAL IN CYLINDRICAL COORDINATES: MOTIVATING QUESTIONS

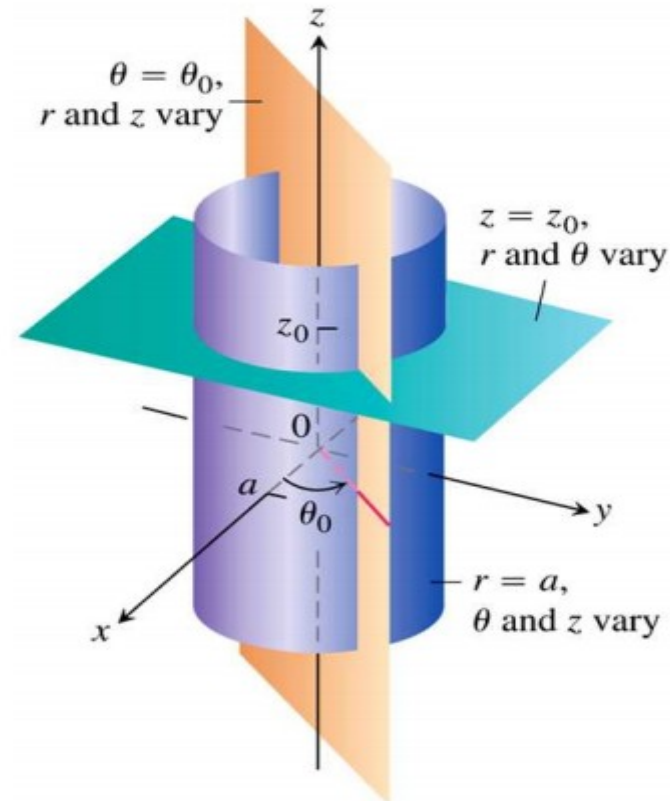
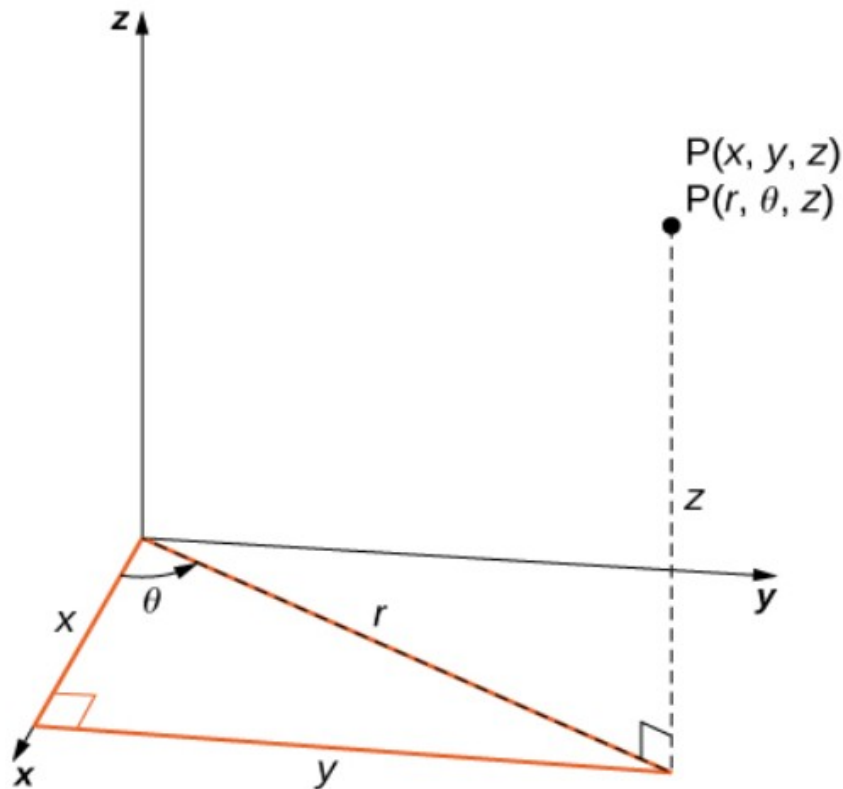


- What are the cylindrical coordinates of a point, and how are they related to Cartesian coordinates?
- What is the volume element in cylindrical coordinates? How does this inform us about evaluating a triple integral as an iterated integral in cylindrical coordinates?

We have encountered two different coordinate systems in $\mathbb{R}^2$ — the rectangular and polar coordinates systems — and seen how in certain situations, polar coordinates form a convenient alternative. In a similar way, there are two additional natural coordinate systems in $\mathbb{R}^3$. Given that we are already familiar with the Cartesian coordinate system for $\mathbb{R}^3$, we next investigate the cylindrical and spherical coordinate systems (each of which builds upon polar coordinates in $\mathbb{R}^2$). In what follows, we will see how to convert among the different coordinate systems, how to evaluate triple integrals using them, and some situations in which these other coordinate systems prove advantageous.

TRIPLE INTEGRAL IN CYLINDRICAL COORDINATES:

In three-dimensional space $\mathbb{R}^3$ a point with rectangular coordinates (x, y, z) can be identified with cylindrical coordinates (r, θ, z) and vice versa.



To convert from rectangular to cylindrical coordinates, we use the conversion

- $x = r \cos \theta$
- $y = r \sin \theta$
- $z = z$

To convert from cylindrical to rectangular coordinates, we use

- $r^2 = x^2 + y^2$ and
- $\theta = \tan^{-1} \left(\frac{y}{x} \right)$
- $z = z$

Note that that z -coordinate remains the same in both cases.

For **cylindrical coordinates**, the change of variables function is

$$(x, y, z) = \mathbf{T}(r, \theta, z)$$

where the components of $\mathbf{T}$ are given by

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z.$$

We can compute that

$$\begin{aligned} D\mathbf{T}(\rho, \theta, \phi) &= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial z} \end{vmatrix} \\ &= \begin{vmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} \\ &= r \cos^2 \theta + r \sin^2 \theta = r. \end{aligned}$$

Note that if $g(x, y, z)$ is the function in rectangular coordinates and the box B is expressed in rectangular coordinates, then the triple integral

$$\iiint_B g(x, y, z) dV$$

is equal to the triple integral

$$\iiint_B g(r \cos \theta, r \sin \theta, z) r dr d\theta dz$$

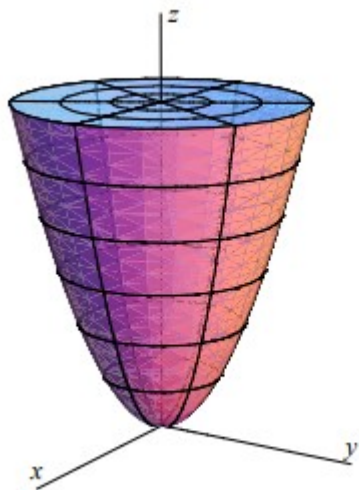
and we have

$$\iiint_B g(x, y, z) dV = \iiint_B g(r \cos \theta, r \sin \theta, z) r dr d\theta dz = \iiint_B f(r, \theta, z) r dr d\theta dz.$$

	Circular cylinder	Circular cone	Sphere	Paraboloid
Rectangular	$x^2 + y^2 = c^2$	$z^2 = c^2(x^2 + y^2)$	$x^2 + y^2 + z^2 = c^2$	$z = c(x^2 + y^2)$
Cylindrical	$r = c$	$z = cr$	$r^2 + z^2 = c^2$	$z = cr^2$

EXAMPLE 1: TRIPLE INTEGRAL IN CYLINDRICAL COORDINATES

Let $\mathcal{U}$ be the solid enclosed by $z = x^2 + y^2$ and $z = 9$. Rewrite the triple integral $\iiint_{\mathcal{U}} x \, dV$ as an iterated integral. (You need not evaluate, but can you guess what the answer is?)



, the given triple integral is equal to the iterated integral

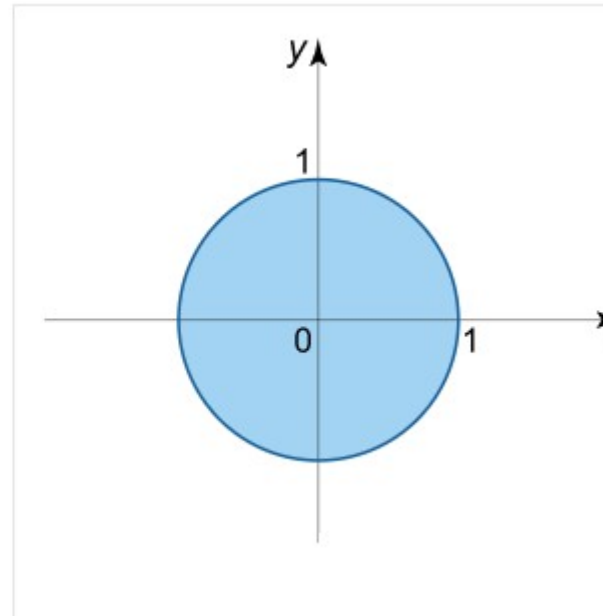
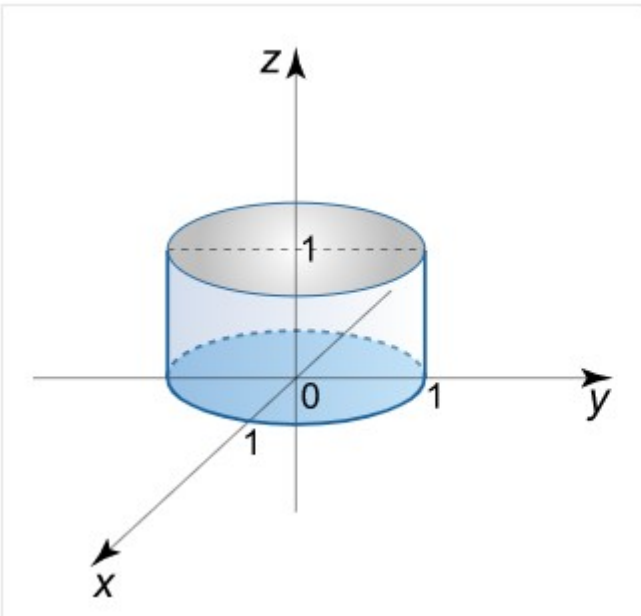
$$\begin{aligned} \boxed{\int_0^9 \int_0^{2\pi} \int_0^{\sqrt{z}} r \cos \theta \cdot r \, dr \, d\theta \, dz} &= \int_0^9 \int_0^{2\pi} \left(\frac{1}{3} r^3 \cos \theta \Big|_{r=0}^{r=\sqrt{z}} \right) dr \, d\theta \, dz \\ &= \int_0^9 \int_0^{2\pi} \frac{1}{3} z^{3/2} \cos \theta \, d\theta \, dz \\ &= \int_0^9 \left(\frac{1}{3} z^{3/2} \sin \theta \Big|_{\theta=0}^{\theta=2\pi} \right) dz \\ &= \boxed{0} \end{aligned}$$

PROBLEMS

Evaluate the integral

$$\iiint_U (x^4 + 2x^2y^2 + y^4) dx dy dz,$$

where the region U is bounded by the surface $x^2 + y^2 \leq 1$ and the planes $z = 0, z = 1$

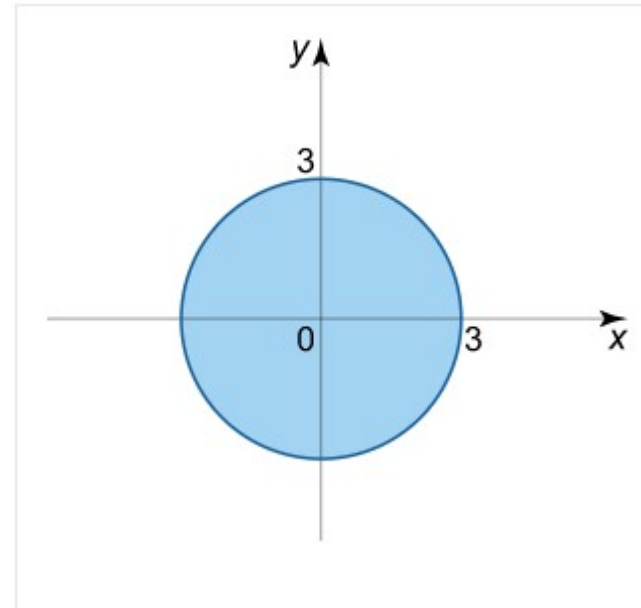
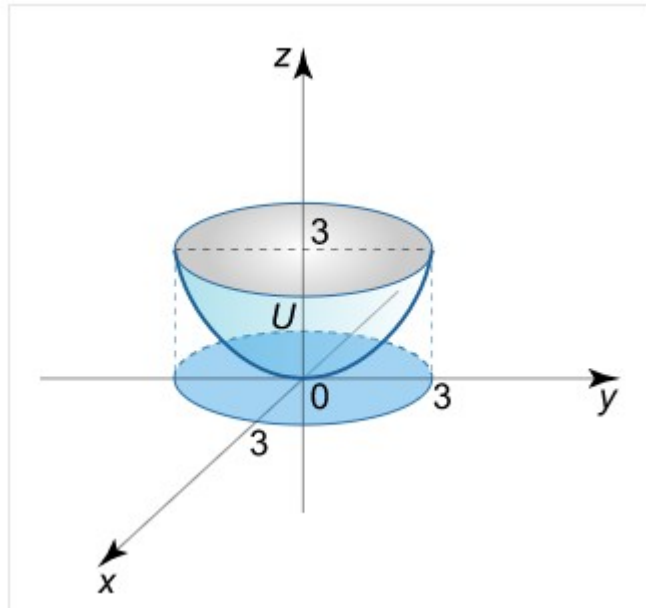


Problems

Find the integral

$$\iiint_U (x^2 + y^2) \, dx \, dy \, dz,$$

where the region U is bounded by the surfaces $x^2 + y^2 = 3z$, $z = 3$.





THANK YOU
